Read-level re-analysis of a child's genome: compound-heterozygous BUB1B variants in mosaic variegated aneuploidy

Track 1 report · Rare Disease, Real Kid: MVA Hackathon 2026
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Submission: `findings/track1/submission_full.csv` (final Track 1 submission) · Code: https://github.com/smarin00/mva-hackathon-2026

Summary. The child's whole-genome reads were re-aligned and re-called independently of the vendor, then prioritised by phenotype with Exomiser. **BUB1B** ranked first by a wide margin. The primary prediction is a compound-heterozygous pair in *BUB1B*: a stop-gain, p.Leu737*, and an extremely rare missense change in the pseudokinase domain, p.Asn1002Lys, submitted together on one row with each allele also listed singly. Biallelic *BUB1B* variants cause mosaic variegated aneuploidy type 1 (MVA1), and one truncating plus one missense allele is the genotype pattern reported in MVA1. DeepVariant recovered both alleles from the reads at germline allele fractions; read-backed phasing could not resolve whether they lie in trans. Read-level checks found no mosaic, structural or repeat-expansion variant in or near *BUB1B*, and no confirmed chromosome-level dosage change. One secondary finding (LZTR1) is reported. The automatic submission rules alone produce the same four variants; a clean rerun of the whole pipeline reproduced the submission byte for byte.

Note. An earlier submission (6 October 2026) ranked the provided vendor VCF with Exomiser alone. This report describes the final, read-level submission.

1. Case and data

The child presented with rhabdomyosarcoma (HPO HP:0002859), the oncological event that triggered the genetic investigation.

Input	Use
Raw reads	Four lanes of paired-end whole-genome FASTQ: the basis of this submission
Phenotype	HPO terms from the clinical phenotype document, each checked against the current HPO release (`pyhpo`)
Vendor VCF	Not used for this submission (basis of the earlier one); both causal alleles are also present in it

2. Pipeline

One Nextflow workflow (`make all` with `mode: full` in the case file `params.yaml`) on GRCh38, no-alt analysis set with hs38d1 decoys.

Step	Tool (version)	Output used
Read QC and trimming	fastp 1.0.1 [1]	per lane
Alignment	bwa-mem 0.7.19 [2], per lane; merge, sort, duplicate marking (samtools)	CRAM
Coverage	mosdepth 0.3.14 [3]	depth QC, aneuploidy
Variant calling	DeepVariant 1.10.0 [4], primary chromosomes	germline SNVs and indels
Normalisation	bcftools: PASS only, primary contigs, multiallelic split, left-aligned	input to ranking
Prioritisation	Exomiser 15.1.0 [5, 6], data 2602 (settings below)	ranked candidates
Phasing	WhatsHap 2.8 [7], candidate-gene panel	phase of candidate pairs
Mosaic variants	GATK 4.7.0.0 Mutect2 [8], candidate-gene panel	low-fraction SNVs and indels
Structural variants	Delly 2.7.0 [9]	SVs
Repeat expansions	ExpansionHunter 5.0.0 [10], GRCh38 catalogue; sex from chrY/chrX depth	STR genotypes
Aneuploidy	depth and B-allele fraction per chromosome, three GC-correction models (`bin/aneuploidy_plot.py`)	per-chromosome dosage
Annotation	Ensembl VEP 116.2 [11] with AlphaMissense [12] and ClinVar 2026-10-04	candidate review

The candidate-gene panel (132 MVA, spindle-checkpoint, kinetochore, centromere, cohesion, APC/C and microcephaly genes) is listed in `docs/panel_genes.txt`.

Exomiser settings. Analysis mode PASS_ONLY; coding and splice-region effects; maximum population frequency 2% (gnomAD v4.1, UK10K); pathogenicity sources REVEL, MVP, AlphaMissense and SpliceAI; all inheritance modes; hiPHIVE (human, mouse and zebrafish phenotypes and protein interactions) and OMIM prioritisers.

3. Submission construction and EPCR

The ranked variants are turned into the submission by a scripted, validated step (`bin/make_submission.py`), not by hand:

- **Compound-heterozygous pairs on one row**, built only from variants that Exomiser scored under a recessive mode; each allele is also listed below the pair as a hedge.
- **Rules applied to every genome:** ClinVar-benign variants excluded; hypervariable loci excluded (HLA, KIR, MUC, olfactory receptors, Ig/TCR); variants less than 50 bp apart are treated as one event, never a pair; lone heterozygous hits in recessive-only genes excluded; Exomiser gene score below 0.1 excluded; population frequency must fit the mode of inheritance (at most 0.1% for dominant, 2% for recessive); genes with no OMIM or Orphanet disease in the HPO annotations are ranked last. Tests on synthetic data check these rules on every push.
- **Documented gene-level review** (`config/cases/mva1_proband.curation.tsv`): the remaining candidates below the leading gene were reviewed, and each was excluded or kept as a secondary finding with its reason. The script applies these decisions, so the file stays reproducible.
- **Secondary findings ranked last**, below every primary row.
- **EPCRs** strictly distinct and descending in rank order; they are **not calibrated probabilities.**
- **Validation** against the scorer's rules, and every variant checked to exist exactly as written in the called VCF.

4. Results

4.1 Exomiser gene ranking

Rank	Gene	Mode	Combined score	Phenotype score
1	BUB1B	AD	0.587	0.564
2	BUB1B	**AR**	0.554	**0.792**
3	LZTR1	AD	0.147	0.510
4	HK1	AD	0.141	0.505
5	HLA-DRB1	AD	0.057	0.684

BUB1B leads by a wide margin. The AD and AR rows describe the same two alleles; the higher phenotype score under AR reflects the match between the child's HPO profile and MVA1, a recessive disorder.

4.2 Submitted predictions

Rank	Row	Gene	EPCR	Consequence	Max. population frequency	ClinVar	Finding
1	pair	BUB1B	0.95	stop-gain + missense	0.0100% + 0.0001%	P/LP + not in ClinVar	primary
2	single	BUB1B	0.70	stop-gain	0.0100%	P/LP	primary (hedge)
3	single	BUB1B	0.45	missense	0.0001%	not in ClinVar	primary (hedge)
4	single	LZTR1	0.01	stop-gain	0.0030%	P/LP	**secondary**

All variants are heterozygous. The automatic rules alone produce these same four variants; the gene-level review changes only LZTR1, from primary to secondary, and removes nothing else from the top rows.

BUB1B. *BUB1B* encodes BubR1, a core component of the spindle assembly checkpoint, and biallelic variants cause MVA1 [13]. The child carries:

- **chr15:40209701 T>G, c.2210T>G, p.Leu737***: stop-gain, very rare (0.01%), pathogenic or likely pathogenic in ClinVar; a loss-of-function allele;
- **chr15:40220612 T>G, c.3006T>G, p.Asn1002Lys**: missense in the pseudokinase domain, extremely rare (0.0001%), not in ClinVar, predicted damaging (AlphaMissense 0.92); a candidate hypomorphic allele.

DeepVariant called both heterozygous from the reads: the stop-gain in 26 of 48 reads (GQ 40) and the missense in 13 of 29 (GQ 35), allele fractions of 0.54 and 0.45, as expected for germline heterozygous variants. Mutect2 found no low-fraction call in *BUB1B*: both alleles are germline, at about 50%.

Phase. The two variants lie 10.9 kb apart, beyond the reach of a read pair, and WhatsHap found no phase block spanning both. Trans configuration is supported by the phenotype (MVA with constitutional aneuploidy requires biallelic loss) but not proven; parental testing or long reads would settle it.

LZTR1, secondary finding. A heterozygous stop-gain, pathogenic or likely pathogenic in ClinVar. Heterozygous LZTR1 loss of function predisposes to schwannomatosis with incomplete penetrance. It is unrelated to MVA and ranked below all primary rows.

The mechanism of the two *BUB1B* alleles, and the drug hypotheses that follow from it, are analysed in the Track 2 report (`docs/smarin00_track2_report.pdf`).

4.3 Supporting read-level analyses

- **Mosaic variants** (Mutect2, candidate panel): 12,752 PASS calls, 57 of them below 20% allele fraction, none in *BUB1B*.
- **Structural variants** (Delly): no confident call in or within 100 kb of *BUB1B*.
- **Repeat expansions** (ExpansionHunter, 38 catalogue loci): all within normal ranges. FMR1 carries 46 CGG repeats, in the intermediate range (45–54): below a premutation and not pathogenic. Sex for X-linked loci was set from chrY/chrX read depth (male).

- **Aneuploidy** (Figure 1): after GC correction, chromosomes 16, 17 and 19 show 1.3–2.3% higher depth, depending on the GC model. Chromosome GC content predicts the shift under all three correction models (Spearman ρ 0.48–0.69), and the B-allele fraction shows no matching signal, so these are reported as unconfirmed. The noise-based detection limit is about 2% of cells. Not seeing MVA aneuploidy here is expected: it differs from cell to cell and averages out in bulk DNA.

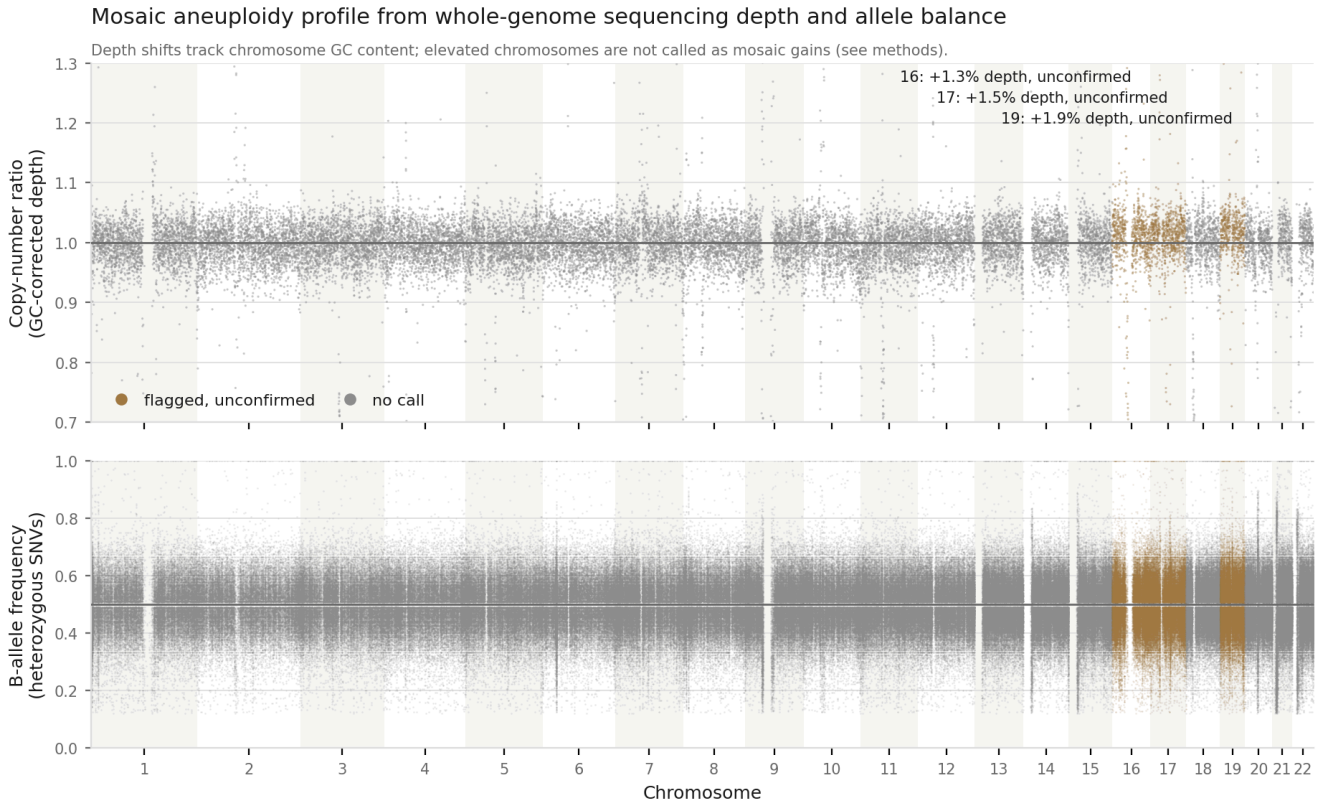


Figure 1. Chromosome dosage from sequencing depth and allele balance. Top: GC-corrected copy-number ratio per bin; bottom: B-allele frequency of heterozygous SNVs. Chromosomes 16, 17 and 19 are flagged but unconfirmed, because their depth shift follows chromosome GC content and the allele balance does not move. Per-chromosome tables for the three GC models: `findings/aneuploidy/` .

5. Limitations

- The phase of the *BUB1B* pair is not established (singleton, no parental data).
- Exomiser considers coding and splice-region variants; deep intronic variants were not systematically scored. SpliceAI was run on the two *BUB1B* alleles only and predicts no splice effect.
- The missense effect is predicted here, not measured; the Track 2 report tests it against experimentally characterised alleles.
- Bulk sequencing cannot measure mosaic aneuploidy at the levels expected in MVA.
- EPCRs are rank-based, not calibrated. The gene-level review involves judgement, documented per gene.

6. Reproducibility

- **One command:** `make all` with `mode: full` in `params.yaml`, about 15 hours on a 16-vCPU / 32 GB cloud server (Hetzner CPX62, about EUR 0.25 per hour), under EUR 5 of compute. One-off setup: about 125 GB of reference data and containers (`make refs`).
- **Reproduced:** a clean clone of the repository, run end to end, reproduced the submission file byte for byte. Per-chromosome aneuploidy estimates agreed within 0.03 percentage points; read placement by bwa-mem varies slightly with thread batching, but the submission did not change.
- **Tested:** 15 tests on synthetic data (`make test`) run on every push (GitHub Actions).
- **Versions:** `docs/reference_data.md` and `findings/versions.txt`; Python environment locked in `requirements.lock`; container checksums recorded. Decisions are logged in `docs/decisions.md`.

7. Data handling and AI use

All analysis ran on a dedicated, key-only cloud server. No genotype-level data is held in the code repository (history checked). All data and genotype-level derived files will be deleted by 23 November 2026 (`docs/DELETION.md`).

AI assistance: Anthropic, Claude via claude.ai, consumer Pro plan (Claude Opus 5.5); model training disabled ("Help improve Claude" off); no outputs rated. It was used for pipeline design, code, troubleshooting and report drafting; all analyses were run and reviewed by the author. No genome-scale data was shared with it (`docs/llm_usage.md`).

References

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